Grid Connected PV System Using Multilevel Inverter

Deepak Kumar Singh
Electrical Engineering Department
NIT Jamshedpur
Jamshedpur, India
dksingh1946@gmail.com

Saibal Manna
Electrical Engineering Department
NIT Jamshedpur
Jamshedpur, India
mannasaibal1994@gmail.com

A.K.Akella
Electrical Engineering Department
NIT Jamshedpur
Jamshedpur, India
akakella.ee@nitjsr.ac.in

***Abstract*—The system is designed to feed the solar energy into a single-phase utility grid. The output frequency and voltage magnitude of the Multilevel Inverter (MLI) is regulated to track the grid frequency and voltage in such a way that Unity Power Factor (UPF) is always maintained. To track the parameters of the grid a Proportional Integral (PI) current controlled algorithm is used. The parameters of the PI control should be selected in such way that the injected grid current should be sinusoidal and UPF along with better dynamic response under continuously changing weather condition. The system is designed with MATLAB/SIMULINK.**

Keywords—Photovoltaic, Multilevel Inverter, Proportional Integral, Grid Connected, Simulink

I. INTRODUCTION

Petroleum, natural gas, and coal are conventional fossil fuel energy sources which fulfill most of the world energy demand and are decreasing day by day. Also, the waste products are great cause of concern such as the greenhouse effect [1]. The Renewable Energy Sources (RES) like wind, tidal, solar, geothermal and biomass are the recent topic for research and development. Among RES, the Photo-voltaic (PV) is widely used as the fabrication of PV devices becomes a reality in low cost [2]. The PV generator converts solar energy into electrical energy. It has many advantages such as pollution free, no moving parts, no wear and tear, free and abundant, and long life span i.e. more than 20 years [3]. As PV generators are green source of energy they are replacing polluting ways of energy production and even more popular for electricity generators [4]. The recent development in power electronics is the MLI. By the use of which, the power conversion is performed with enhanced power quality having less Total Harmonic Distortion (THD), less Electromagnetic Interface (EMI) and require low filter size [5]. Hence, it becomes popular and demanding due to its advantages over conventional inverter. Besides that it has capability to handle higher power and higher voltages along with less voltage stress on the power electronics switch [6].

II. PV SYSTEM EQUIVALENT CIRCUIT

The electric circuit presented in Fig. 1. represents the PV cell. The following equations show the current voltage relationship. Where I_P is the photo current that is directly proportional to the incident solar radiation and terminal current (I) of PV cell is obtained from the equivalent circuit.

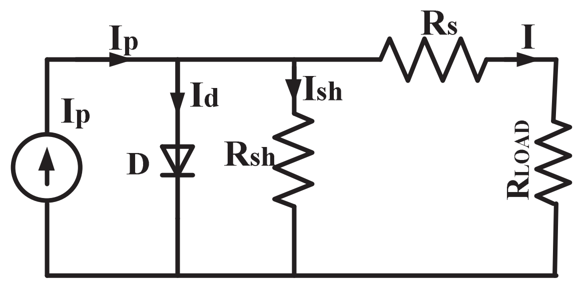

Fig. 1. PV system equivalent circuit

$$I_P = I_d + I_{sh} + I \tag{1}$$

$$I = I_P - I_d - I_{sh} \tag{2}$$

$$I = I_P - I_O\left(e^{\frac{V_{oc}+IR_s}{nV_T}} - 1\right) - I_{sh} \tag{3}$$

$$V_T = \frac{kT}{q} \tag{4}$$

R_{sh} and R_s are shunt and series resistance of PV system. The origin of R_{sh} seems to be more difficult to describe and pertaining to the non-ideal nature and impurities close to the cell's ends which result a short-circuit path near this junction. Ideally, R_s will be null and R_{sh} will be infinite. However, this ideal condition is not feasible and the effect of both resistors is minimized to boost their product. I_d is the current from the diode which is come from the p-n junction theory. T, k and q are temperature, Boltzmann constant and electron charge respectively. I_0 is reverse saturation current and it depends on doping of p-n junction and temperature.

n = 2 for silicon and it depends on material.

In current-voltage characteristic of PV cell, there are two significant points V_{oc} i.e. open circuit voltage and I_{sc} i.e. short circuit current. The power produced at both points is zero. When PV cell output current is zero and R_{sh} is ignored, (5) is approximated and hence, find the approximated value of V_{oc} in (6). The I_{sc} is the current at which output voltage of the circuit is zero and is approximately equal to I_P i.e., the photo current as shown in (7).

$$V_{OC} = nV_T \ln\left(\frac{I_P + I_O}{I_O}\right) \quad (5)$$

$$V_{OC} \propto \ln(I_P) \quad (6)$$

$$I_{SC} \propto I_P \quad (7)$$

When the product i.e. VI is maximum. Then, the PV system produces maximum power. The point can be recognized as a Maximum Power Point (MPP) which is distinctive.

III. MAXIMUM POWER POINT TRACKING (MPPT) ALGORITHM

MPPT techniques are employed to generate maximum power output by tracking MPP in PV system. It basically decides the duty cycle (α) of the converter (for distinct insolation and temperature). For which converter can withdraw maximum power from PV cell.

A. Perturbation and Observation method (P & O)

P&O is frequently used technique for MPPT. P&O technique is employed since it is considered as the best and simple method. The P&O approach creates a disruption in the working voltage of the grid and adjusts the panel voltage by changing the α of the converter. Looking at Fig. 2. i.e. the P&O algorithm flow chart clears the concept behind its operation. If α is decreased on right side of the MPPT then power will increase. Again, if α is increased on left side of the MPPT then power will increase.

B. Boost Converter with MPPT

The complete block diagram of a Boost converter with MPPT is illustrated in Fig. 3. In this PV module, Boost converter and MPPT are connected to the load. The MPPT detects the current and voltage outcome from the PV cell and decides the α for which the boost converter helps to withdraw the maximum power from the PV module. A controls the PWM signals that help to trigger the power electronic switch used in boost converter. So, the converter experiences a particular load for which the PV module gives the maximum power to the load.

IV. MLI AND ITS PULSE WIDTH MODULATION TECHNIQUES

The PWM inverters will simultaneously regulate the output voltage and frequency. And harmonics in load currents can also be minimized. However, a significant interest in reducing the effect of EMI or noise and vibrations remains the reduction of harmonically components in the current. Many MLI have been presented last year that synthesis a great number of levels [7]. There are also many benefits in approaches in these technical papers, this includes an enhanced output, a smaller filter size, a smaller EMI and other advantages.

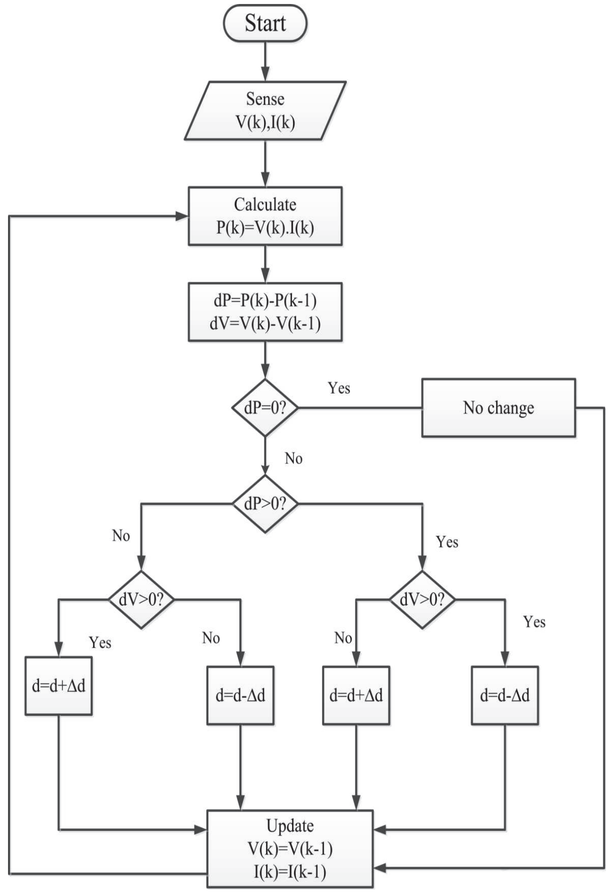

Fig. 2. Flow chart of P&O algorithm

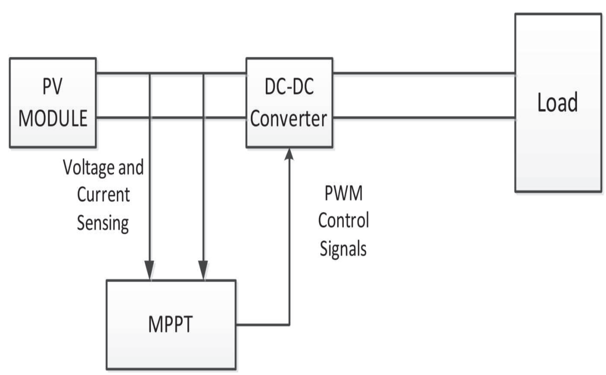

Fig. 3. Overall system Block Diagram

A. Nine Level Inverter Operation

The circuit configuration for nine level inverter is comprising of two five level inverter connected in cascade. That means, two auxiliary circuits and two full bridge configurations are the main parts for the 1-ϕ nine level inverter as displayed in Fig. 4.

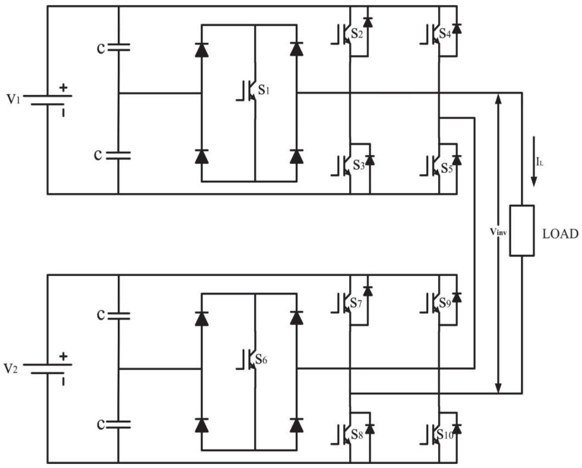

Fig. 4. Single phase nine level inverter

B. Working principle of PWM Technique

This inverter topology uses a dual references phase shift modulation technique to obtain driven pulse signals for the power electronics switches. In this modulation technique two references signals V_{ref1} and V_{ref2} as well as two identical triangular carrier signals $V_{carrier1}$ and $V_{carrer2}$ are used as shown in Fig. 5. Here, $V_{carrer2}$ is phase shifted by half of the time period of $V_{carrier1}$ and the two reference signals V_{ref1} and V_{ref2} are similar to each other, only difference for the offset value which is equal to the peak magnitude of the carrier signal. The switching pattern for nine level inverter is displayed in Table I.

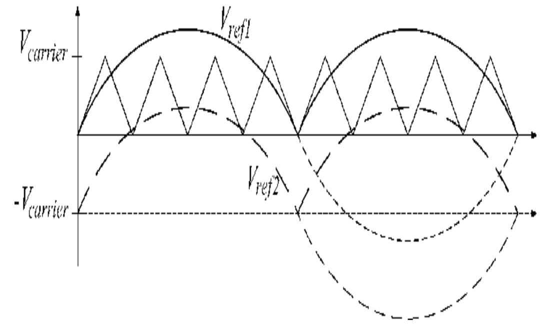

Fig. 5. Reference signal and carrier signal

C. CONTROL OF NINE LEVEL INVERTER FOR GRID CONNECTED PHOTO-VOLTAIC SYSTEM

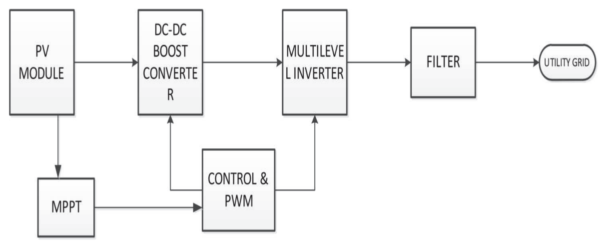

Fig. 6. Block diagram of PV system synchronised with utility grid

The complete illustration of grid connected PV system is displayed in Fig. 6. The suggested 1-ϕ nine level inverter comprising of two modules, in cascaded manner and are connected to the utility grid as represented in Fig. 7., where each module is comprised of a Boost converter, PV array, auxiliary circuit with full H-Bridge configuration. Boost converter, connected after the PV array, is utilized to withdraw maximum power from the PV array and also to get controlled step up output voltage to inject only active component of current to the utility grid.

TABLE I. SWITCHING TABLE OF 9 LEVEL INVERTER

Vinv	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
2VDC	OFF	ON	OFF	OFF	ON	OFF	ON	OFF	OFF	ON
1.5VDC	OFF	ON	OFF	OFF	ON	ON	OFF	OFF	OFF	ON
VDC	OFF	ON	OFF	OFF	ON	OFF	OFF	OFF	OFF	ON
0.5VDC	ON	OFF	OFF	OFF	ON	OFF	OFF	OFF	OFF	ON
0	OFF	OFF	OFF	OFF	ON	OFF	OFF	OFF	OFF	ON
0	OFF	OFF	ON	OFF	OFF	OFF	OFF	ON	OFF	OFF
-0.5VDC	ON	OFF	OFF	ON	OFF	OFF	OFF	OFF	ON	OFF
-VDC	OFF	OFF	ON	ON	OFF	OFF	OFF	OFF	ON	OFF
-1.5VDC	OFF	OFF	ON	ON	OFF	ON	OFF	OFF	ON	OFF
-2VDC	OFF	OFF	ON	ON	OFF	OFF	OFF	OFF	OFF	ON

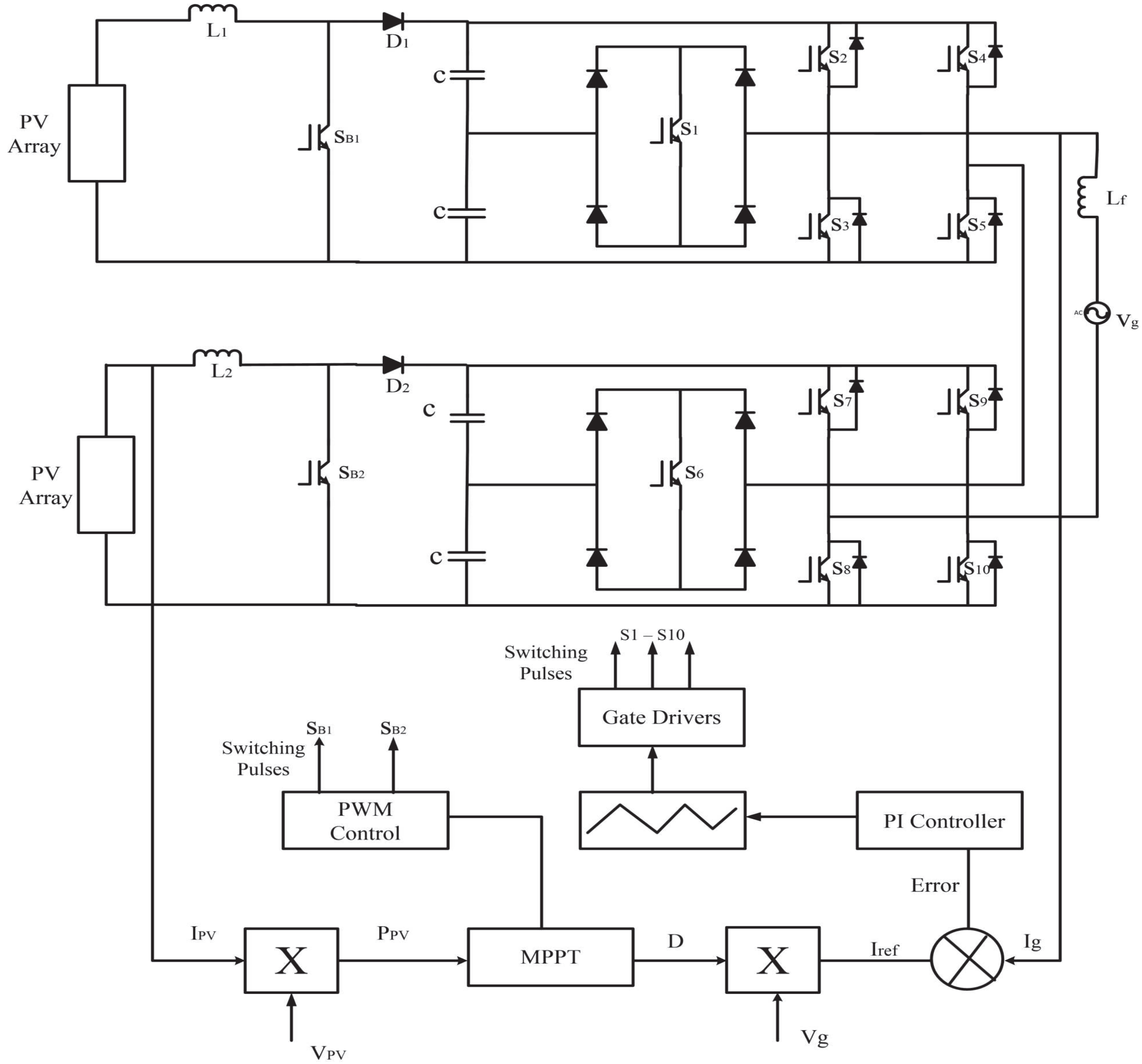

Fig. 7. Complete circuit configuration along with the control scheme

Because the condition for the current to be flown from inverter to grid is the inverter voltage (V_{inv}) should be greater than $\sqrt{2}$ times of the utility grid voltage (V_g).

$$V_{inv} > \sqrt{2} V_g \tag{8}$$

To filter out the harmonic components, a filter having inductance, L_f is employed as current injected to the utility grid should be sinusoidal in nature and contain small amount total harmonic distortion.

V. Result

Simulation has been done by using MATLAB/SIMULINK. The characteristics curves and the result are taken into consideration for verification. By varying the circuit parameter complete topology has been analyzed. The simulation parameters are here taken sampling time (T_S) = 20 μs and the fixed step size = 1 μs , carrier frequency (f_c) = 20 kHz.

A. PV Array

A PV Array has been modeled from its mathematical equivalent circuit by using the software and the specifications of PV module are displayed in Table II and the output is shown in Fig. 8. and Fig. 9.

Table II. Parameters of PV Module

Description	Rating
Nominal power	49.80 W
Maximum power point voltage	22.1 V
Maximum power point current	2.25 A
V_{oc}	25.8 V
I_{sc}	2.95 A
Total series connected cells	36
Total parallel connected cells	1

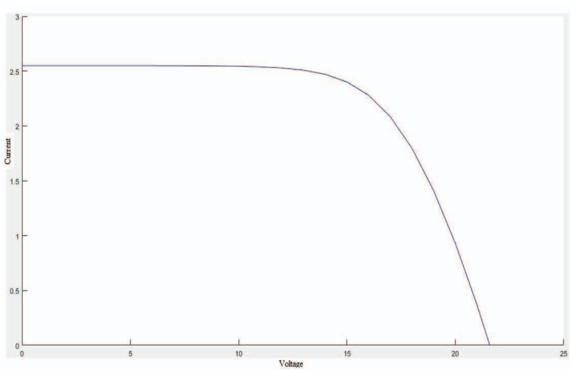

Fig. 8. Simulation result of I-V characteristics for a PV module

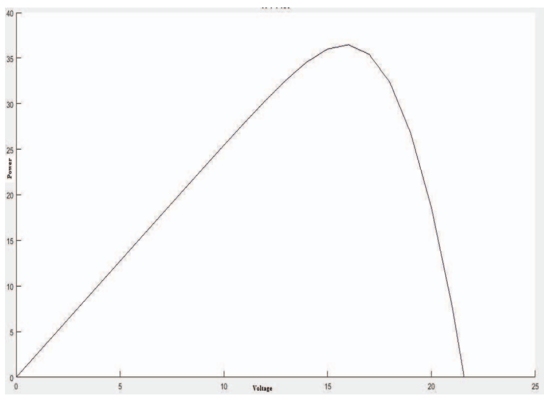

Fig. 9. Simulation result of P-V characteristics for a PV module

B. Simulation Result for the proposed topology

The simulation result of inverter output and grid input for modulation index (M<1) taking reference signal frequency as 50 Hz and carrier frequency as 20 kHz as displayed in Fig. 10. and Fig. 11. The parameters of various components are shown in Table III.

TABLE III. PARAMETERS OF VARIOUS COMPONENTS

PARAMETERS	VALUE
Kp	10
Ki	0.1
Carrier frequency	20 kHz
Sampling Time (Ts)	20 µs
Filter (L_f)	1 mH
C_1-C_4	2200 µF

Keeping the condition $V_{inv} > \sqrt{2}V_g$, from simulation results, it is cleared that for M<1, the inverter output voltage levels are leveled correctly and the current into the utility grid is almost sinusoidal wave in nature. Hence, it experiences low value of THD and its value lies within 5% after the filter. Also, it has been observed from the result that the power factor is nearly to unity. If M >1 then the inverter output voltage levels are not good.

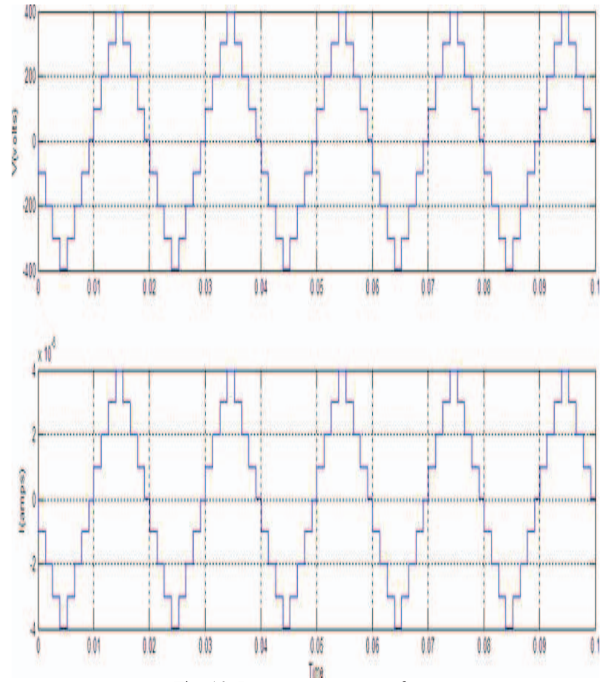

Fig. 10. Inverter output waveform

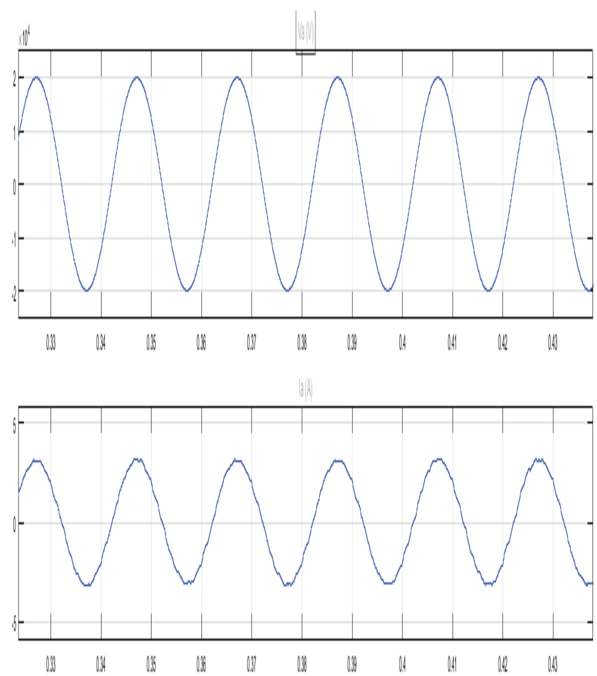

Fig. 11. Grid voltage and current with filter

VI. CONCLUSION

A 1-ϕ nine-level PWM inverter for grid connected PV system has been simulated. The modulation and control of the above MLI topology are based on a generalized novel hybrid dual reference Phase Shifted PWM technique. A 1-ϕ nine-level inverter has been employed in PV system synchronized to utility grid. The DC link voltage across each inverter module has been boosted up to a required value from the suitable number of solar PV arrays using a conventional boost converter. The α of the converter has been generated from the MPPT algorithm. The P&O scheme has been adopted for the extraction of maximum solar energy. The system is designed to feed the solar energy into a single phase utility grid. The output frequency and voltage magnitude of the above MLI has been regulated to track the grid frequency and voltage in such a way that UPF is always maintained. To track the above control parameters of the grid a PI current controlled algorithm has been used. The parameters of the PI control have been selected in such way that, the injected grid current should be sinusoidal and UPF along with better dynamic response under continuously changing weather condition. The above simulation results have been verified with the theoretical approach successfully and concluded to be implemented for experimental purpose.

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